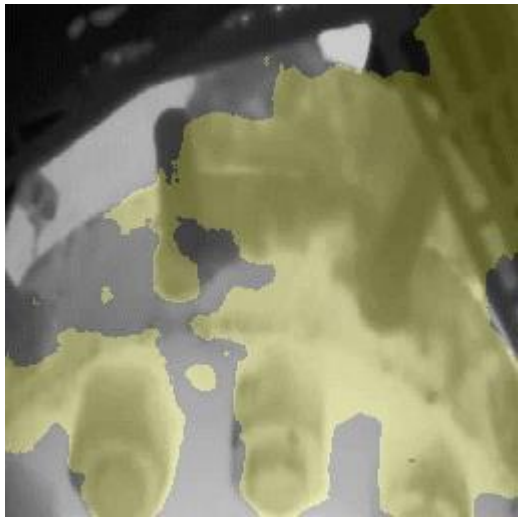
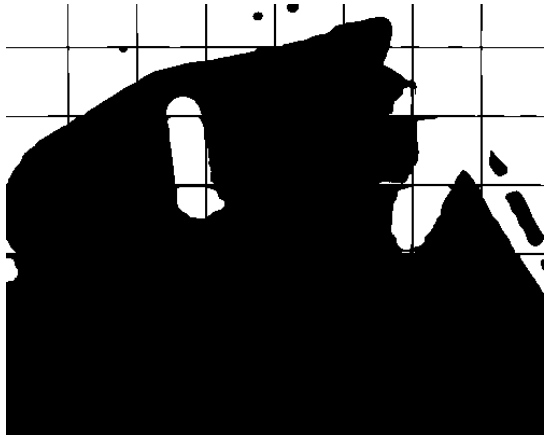
袁



Cosnet

3Dunet



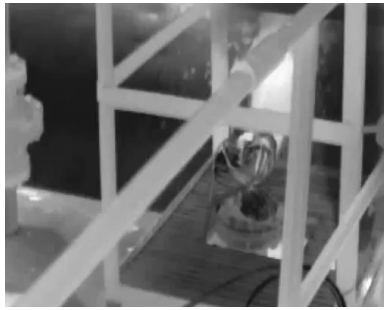
用2D卷积无法捕捉烟羽的流动特性

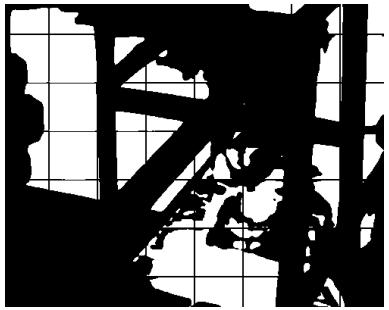
归一化

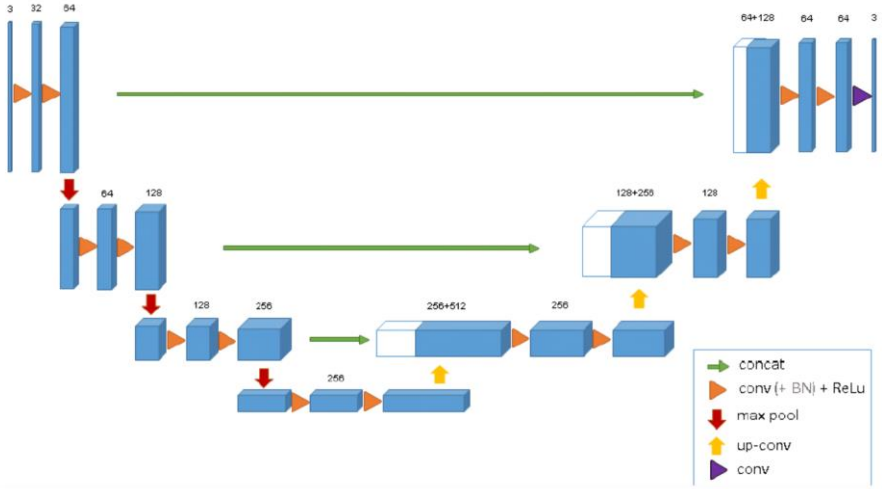
2. 灰度变换归一化*[2]*

灰度变换归一化是利用灰度拉伸的方法将原图像中的灰度分布扩展到具有整个灰度级的图像。例如，采集到的手背静脉图像是8位的灰度图像，应有256个灰度级，但往往由于采集时光照等因素的影响，图像的灰度级集中在一个或者几个灰度级区段，这时可以采用灰度拉伸的方法将图像扩展到256个灰度级。灰度变换归一化的公式为：

$$\mathrm{N(i,j)} = \frac{I(i,j) - min}{max - min} * 255$$






3
32
64
64
128
128
256
256
256
256+512
256
128+256
128
64+128
64
64
3
concat
conv (+ BN) + ReLu
max pool
up-conv
conv

Learning Unsupervised Video Object Segmentation through Visual Attention

Wenguan Wang*[1,2], Hongmei Song *[1], Shuyang Zhao [1],
Jianbing Shen† [1,2], Sanyuan Zhao [1], Steven Hoi [3], Haibin Ling [4]
[1]Beijing Lab of Intelligent Information Technology, School of Computer Science, Beijing Institute of Technology, China
[2]Inception Institute of Artificial Intelligence, UAE [3]Singapore Management University, SGP [4]Temple University, USA
wenguanwang.ai@gmail.com, songhongmei@bit.edu.cn, zsyprich@bit.edu.cn

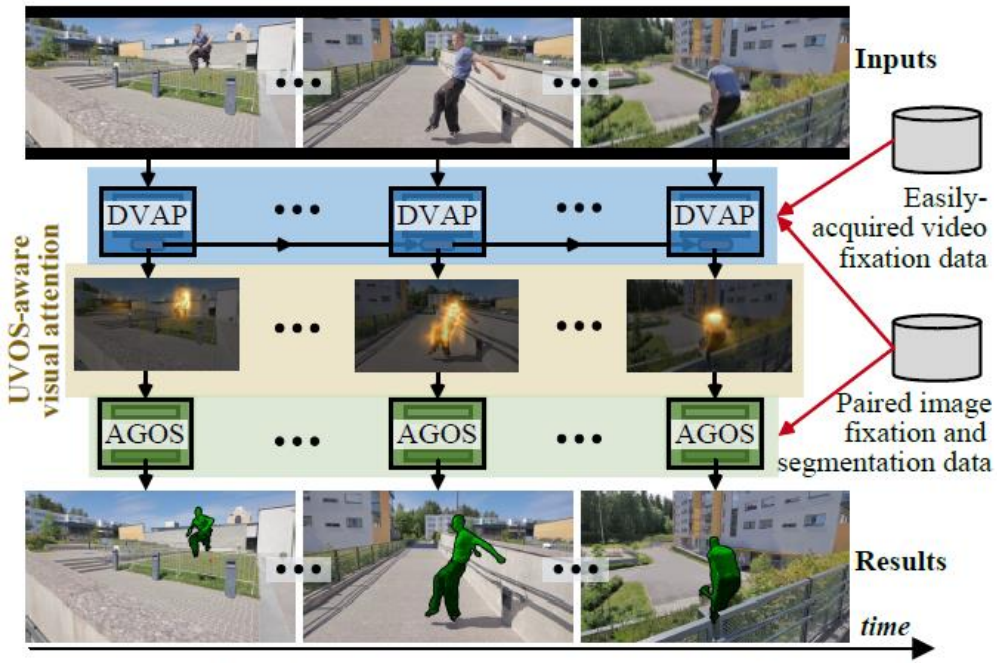


Figure 1. Our UVOS solution has two key steps: Dynamic Visual Attention Prediction (DVAP, §5.2) cascaded by Attention-Guided Object Segmentation (AGOS, §5.3). The UVOS-aware attention from DVAP acts as an intermediate video object representation, freeing our method from the dependency of expensive video object annotations and bringing better interpretability.

we decouple UVOS into two sub-tasks: UVOS-driven Dynamic Visual Attention Prediction (DVAP) in spatiotemporal domain, and Attention-Guided Object Segmentation (AGOS) in spatial domain.

In our UVOS model, the DVAP module is built upon a CNN-convLSTM architecture, where the convLSTM takes static CNN feature sequence as input and learns to capture the dynamic visual attention, and the AGOS module is based on an FCN architecture.

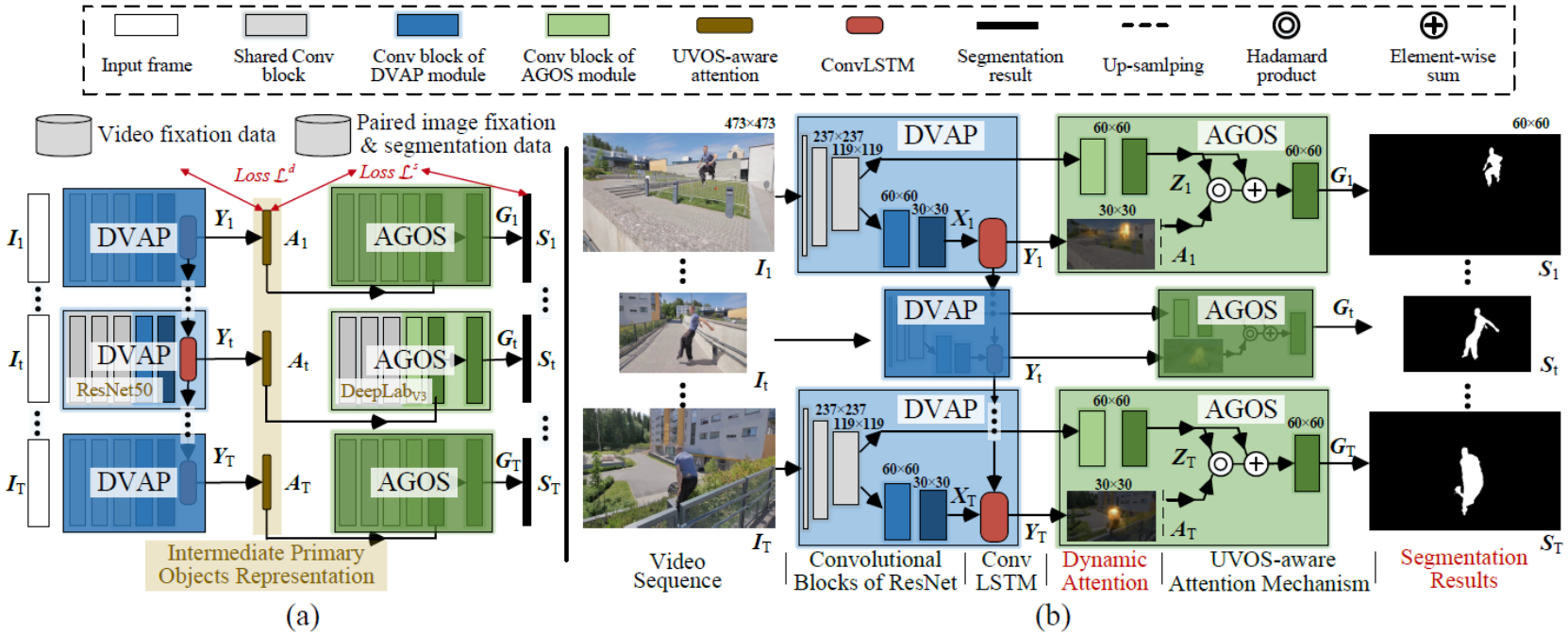


Figure 3. **Illustration of the proposed UVOS model.** (a) Simplified schematization of our model that solves UVOS in a two-step manner, without the need of training with expensive precise video object masks. (b) Detailed network architecture, where the DVAP (§5.2) and AGOS (§5.3) modules share the weights of two bottom conv blocks. The UVOS-aware attention acts as an intermediate object representation that connects the two modules densely. Best viewed in color. Zoom in for details.

总体结果：

$$\{\mathbf{A}_t\}_{t=1}^{T} = \mathcal{F}_{\text{DVAP}}(\{\mathbf{I}_t\}_{t=1}^{T}).$$

$$\mathbf{S}_t = \mathcal{F}_{\text{AGOS}}(\mathbf{I}_t, \mathbf{A}_t), \quad t \in \{1, 2, \ldots, T\}.$$

DVAP Module：

$$\mathbf{X}_t = \text{CNN}(\mathbf{I}_t), \mathbf{Y}_t = \text{convLSTM}(\mathbf{X}_t, \mathbf{Y}_{t-1}), \mathbf{A}_t = \mathcal{R}(\mathbf{Y}_t), \quad (4)$$

where Y_t indicates the 3D-tensor hidden state (with 32
channels) of convLSTM at time step t. R is a readout function
that produces the attention map from the hidden state

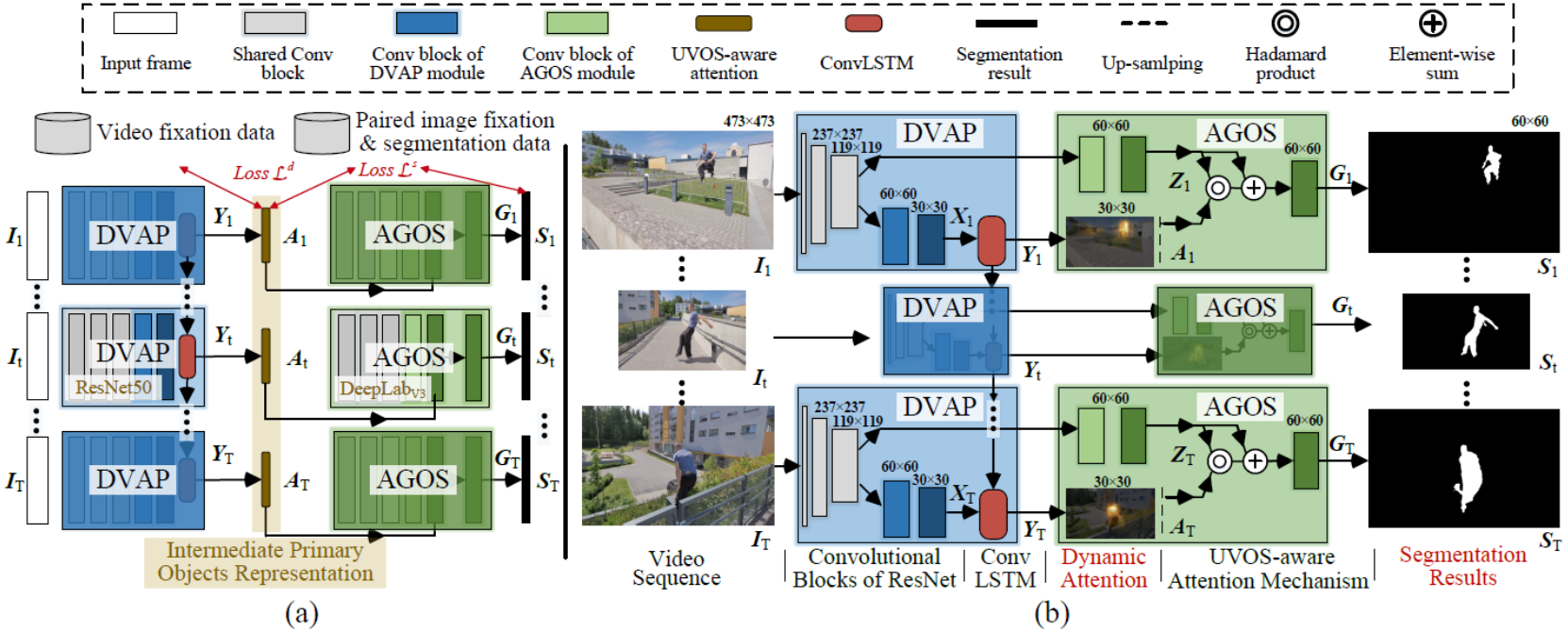


Figure 3. **Illustration of the proposed UVOS model.** (a) Simplified schematization of our model that solves UVOS in a two-step manner, without the need of training with expensive precise video object masks. (b) Detailed network architecture, where the DVAP (§5.2) and AGOS (§5.3) modules share the weights of two bottom conv blocks. The UVOS-aware attention acts as an intermediate object representation that connects the two modules densely. Best viewed in color. Zoom in for details.

AGOS Module

$$\textit{Spatiotemporal attention: } \{\mathbf{A}_t\}_{t=1}^{T} = \mathcal{F}_{\text{DVAP}}(\{\mathbf{I}_t\}_{t=1}^{T}),$$

$$\textit{Spatial feature enhancement: } \begin{aligned} \mathbf{Z}_t &= \mathcal{F}_{\text{ASPP}}(\mathbf{I}_t), \\ \mathbf{G}_t^c &= (1 + \mathbf{A}_t) \odot \mathbf{Z}_t^c. \end{aligned} \tag{8}$$

This strategy retains the original information
(even with a
very small attention value

损失函数

Training loss: For DAVP, given an input frame $\mathbf{I} \in R^{473\times473\times3}$, it predicts an attention map $\mathbf{A} \in [0,1]^{30\times30}$. Denote by $\mathbf{P} \in [0,1]^{30\times30}$ and $\mathbf{F} \in \{0,1\}^{30\times30}$ the ground-truth continuous attention map and the binary fixation map, respectively. $\mathbf{F}$ is a discrete map, recording whether a pixel receives human-eye fixation position, and $\mathbf{P}$ is obtained by blurring $\mathbf{F}$ with a small Gaussian filter. Inspired by [26], the loss function $\mathcal{L}_{\text{DVAP}}$ for DAVP is designed as:

$$\begin{aligned}\mathcal{L}_{\text{DVAP}}(\mathbf{A},\mathbf{P},\mathbf{F}) =& \mathcal{L}_{\text{CE}}(\mathbf{A},\mathbf{P}) + \alpha_1\mathcal{L}_{\text{NSS}}(\mathbf{A},\mathbf{F}) + \\ & \alpha_2\mathcal{L}_{\text{SIM}}(\mathbf{A},\mathbf{F}) + \alpha_3\mathcal{L}_{\text{CC}}(\mathbf{A},\mathbf{P}),\end{aligned} \tag{9}$$

For AGOS, given $\mathbf{I}$, it produces the final segmentation prediction[2] $\mathbf{S} \in [0,1]^{60\times60}$. Let $\mathbf{M} \in \{0,1\}^{60\times60}$ denote the ground-truth binary segmentation mask, the loss function $\mathcal{L}_{\text{AGOS}}$ of the AGOS module is formulated as:

$$\mathcal{L}_{\text{AGOS}}(\mathbf{S},\mathbf{M}) = \mathcal{L}_{\text{CE}}(\mathbf{S},\mathbf{M}). \tag{10}$$

新增了标注

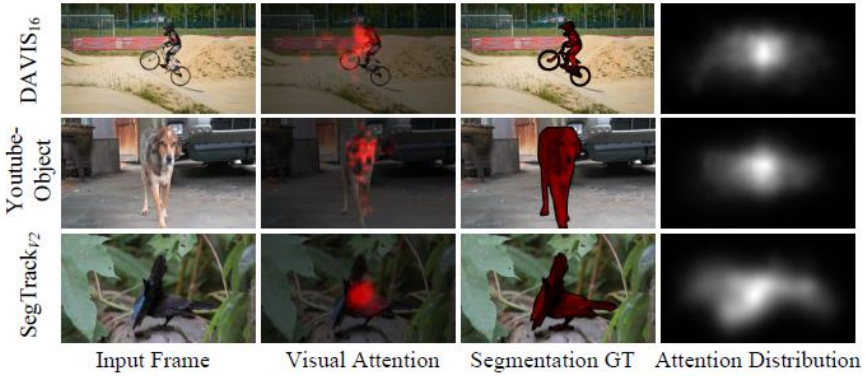


Figure 2. **Example frames from three datasets** ([55, 57, 42]) with our eye-tracking annotation (§3). The last column shows the average attention maps of these datasets. We quantitatively verify (§4) the high consistency between human attention behavior (2nd column) and primary-object determination (3rd column).

实验结果

Dataset	Metric	**Ours**	PDB [60]	ARP [38]	LVO [63]	FSEG [31]	LMP [62]	SFL [11]	FST [53]	CUT [36]	NLC [13]	MSG [50]	KEY [41]	CVOS [61]	TRC [16]
$DAVIS_{16}$	Mean ↑	**79.7**	**77.2**	**76.2**	75.9	70.7	70.0	67.4	55.8	55.2	55.1	53.3	49.8	48.2	47.3
	$\mathcal{J}$ Recall ↑	**91.1**	**90.1**	**91.1**	89.1	83.5	85.0	81.4	64.9	57.5	55.8	61.6	59.1	54.0	49.3
	Decay ↓	**0.0**	0.9	7.0	**0.0**	1.5	1.3	6.2	**0.0**	2.2	12.6	2.4	14.1	10.5	8.3
	Mean ↑	**77.4**	**74.5**	70.6	**72.1**	65.3	65.9	66.7	51.1	55.2	52.3	50.8	42.7	44.7	44.1
	$\mathcal{F}$ Recall ↑	**85.8**	**84.4**	**83.5**	83.4	73.8	79.2	77.1	51.6	61.0	51.9	60.0	37.5	52.6	43.6
	Decay ↓	**0.0**	**-0.2**	7.9	**1.3**	1.8	2.5	5.1	2.9	3.4	11.4	5.1	10.6	11.7	12.9
	$\mathcal{T}$ Mean ↓	44.5	29.1	39.3	**26.5**	32.8	57.2	28.2	36.6	27.7	42.5	30.2	**26.9**	**25.0**	39.1

COSNet

	Method	TRC [17]	CVOS [51]	KEY [31]	MSG [40]	NLC [14]	CUT [9]	FST [42]	SFL [28]	LMP [52]	FSEG [24]	LVO [53]	ARP [30]	PDB [49]	**COSNet**
$\mathcal{J}$	Mean	47.3	48.2	49.8	53.3	55.1	55.2	55.8	67.4	70.0	70.7	75.9	76.2	77.2	**80.5**
	Recall	49.3	54.0	59.1	61.6	55.8	57.5	64.9	81.4	85.0	83.0	89.1	91.1	90.1	**94.0**
	Decay	8.3	10.5	14.1	2.4	12.6	2.2	**0.0**	6.2	1.3	1.5	**0.0**	7.0	0.9	**0.0**
$\mathcal{F}$	Mean	44.1	44.7	42.7	50.8	52.3	55.2	51.1	66.7	65.9	65.3	72.1	70.6	74.5	**79.4**
	Recall	43.6	52.6	37.5	60.0	61.0	51.9	51.6	77.1	79.2	73.8	83.4	83.5	84.4	**90.4**
	Decay	12.9	11.7	10.6	5.1	11.4	3.4	2.9	5.1	2.5	1.8	1.3	7.9	**-0.2**	0.0
$\mathcal{T}$	Mean	39.1	**25.0**	26.9	30.2	42.5	27.7	36.6	28.2	57.2	32.8	26.5	39.3	29.1	31.9